

Original Article

Haplotype analysis of the RAGE gene: identification of a haplotype marker for diabetic nephropathy in type 2 diabetes mellitus

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Abstract

Background. Diabetic nephropathy (DN) represents a devastating complication of diabetes. Family clustering, heterogeneity in the onset and progression and results of segregation studies indicate that susceptibility to DN is a complex trait.

Methods. Common single nucleotide polymorphisms in the RAGE (receptor of advanced glycation end-products) gene (−429T/C, −374T/A, G82S, 1704G/T, 2184A/G and 2245G/A) were studied in the association study comprising 605 Caucasian subjects by means of haplotype analysis in order to identify an eventual haplotype marker for DN in type 2 diabetes. Haplotypes were constructed computationally; frequencies were compared among groups of subjects with type 2 diabetes (DM) and DN, diabetics without DN and non-diabetics. Survival analysis was carried out to ascertain whether certain RAGE haplotypes influence onset of DN in type 2 diabetics.

Results. Significant differences in haplotype frequencies among DM + DN vs DM non-DN and non-DM groups were found ($P=0.0007$ and 0.0013 , respectively; permutation test). Frequency of the RAGE₂ haplotype containing minor alleles in positions −429 and 2184 (CTGGGG) in the DN group was significantly higher than in the two control groups (21.7% vs 12.8% and 13.8%, both $P_{\text{corr}} < 0.003$; two-tail Fisher exact test); odds ratios 1.65 [95% confidence interval (CI): 1.08–2.50; $P=0.020$] and 1.79 (95% CI: 1.22–2.62; $P=0.003$), respectively. In survival analysis, duration of diabetes until the onset of DN (e.g. appearance of persistent proteinuria) was significantly different among RAGE₂ diplotype groups ($P < 0.05$); median DN-free interval was 9.6 years in RAGE₂ +/+ homozygotes, 15.2 years in +/- heterozygotes and 17.0 years in the -/- combination.

Conclusions. The RAGE₂ haplotype is associated with DN in type 2 diabetics and with earlier DN onset and, thus, can be regarded a marker for DN.

Keywords: advanced glycation end-products; complications; diabetic nephropathy; haplotype analysis; polymorphism; RAGE

Introduction

Diabetic nephropathy (DN) develops as a result of haemodynamic and metabolic changes accompanying diabetes. Most of the hyperglycaemia-driven molecular alterations responsible for the development of diabetic complications in general contribute to some degree to the development of DN. The implication of glycoxidation leading to formation of advanced glycation end-products (AGEs) in the pathogenesis of DN is well established [1,2]. AGEs contribute to diabetic tissue injury by both receptor-independent (direct cross-linking of extracellular matrix proteins and modification of intracellular proteins) and receptor-dependent pathways. The latter process is mediated through the interaction with specific cell-surface receptors for AGEs (macrophage scavenger receptor type II, galectin-3, OST-48, 80K-H, CD36 and RAGE). Unlike other receptors, whose function is predominantly scavenging, RAGE possesses a potent signalling role. By virtue of its ability to activate several intracellular pathways leading to the activation of redox-sensitive transcription factor nuclear factor- κ B (NF κ B), and subsequent NF κ B-mediated expression of a number of genes responsible for the alteration of cellular phenotype and function, RAGE actively participates in the processes amplifying tissue injury and sustained inflammation in diabetes [3].

In the human kidney, likewise in rodent experimental models, RAGE is selectively expressed in podocytes but not in mesangial cells or glomerular endothelium [4].

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Although RAGE has been immunolocalized almost exclusively on podocytes (in both normal and diabetic kidneys), tubular changes at least partly induced by AGE–RAGE interaction were indirectly documented by blockade of AGE-mediated NFκB activation by soluble RAGE or anti-RAGE antibodies [5]. In rodent models, RAGE significantly contributes to glomerular pathology — enhanced permeability, inflammation, mesangial expansion and glomerular basement membrane thickening [6–8]. Engagement of RAGE by AGEs (and probably also S100/calgranulins) in diabetic kidney contributes through the cascade of signalling events using reactive oxygen species (ROS) as second messengers to the activation of TGF-β, CTGF and VEGF axes directly responsible for renal remodelling. Therapeutic blockade of AGEs, RAGE or the absence of the latter in RAGE-null mice completely suppressed structural and functional changes associated with DN in mice [6,9], thereby supporting the pathogenic role of RAGE in DN.

Sequence variation within the RAGE gene has been studied and a relatively large number of single nucleotide polymorphisms (SNPs) in coding and non-coding regions of the RAGE gene have been identified recently (Figure 1). The functional impact of several of them on the transcriptional activity [10], ligand binding [11] or intermediate phenotype [12] has been described. Several association studies investigated association of selected SNPs in RAGE with complications of diabetes other than DN, mainly with diabetic retinopathy; however, the majority led to negative results. So far, only a limited number of association studies investigated the relationship between RAGE polymorphisms and DN. Selected SNPs (–429T/C, –374T/A and G82S) were studied in type 1 diabetic subjects with DN in the Finnish population; the results indicated a possible protective role of the –374AA genotype for the development of proteinuria and coronary heart disease (CHD) in type 1 diabetes [13]. Japanese authors presented their finding of an association of the 1704G/T RAGE polymorphism together with NAD(P)H p22phox C242T variant with DN; however, this was in a rather limited sample of type 2 diabetic patients [14]. Recently, the 63 bp insertion/deletion variant in the RAGE promoter has been associated with DN in the German population [15].

With respect to the above-mentioned findings, the RAGE gene is an obvious functional candidate for DN. For the purpose of a future larger multilocus analysis, including among others RAGE, we performed genotyping of all common SNPs in the RAGE gene in order to find a relevant haplotype-tagging SNP (htSNP) and eventual marker for DN. One can presuppose that such a short chromosomal segment (spanning ~4 kb) harbours tightly linked markers and we might be able to detect htSNPs among them, which could be further tested for association with DN. The aims of our study were to perform SNP-based haplotype analysis of the RAGE gene and, consequently, to try to identify an eventual haplotype marker for DN in type 2 diabetes. Moreover, with respect to the prevalence and incidence data on DN, we performed survival analyses to ascertain whether RAGE SNPs (or haplotypes) influence the onset of DN in type 2 diabetics.

Subjects and methods

Subjects

A total of 605 Caucasian subjects (265 males and 340 females) comprising three groups were enrolled in the association study: (i) type 2 diabetics with parallel DN (DM + DN); (ii) type 2 diabetics without DN (DM non-DN); and (iii) non-diabetic subjects (non-DM). Clinical characteristics of the study subjects are shown in Table 1. Diabetic patients were recruited from diabetic outpatient clinics; those with DN were all followed in the specialized nephrology unit of the University Hospital Brno-Bohunice. Non-diabetic subjects were recruited from clients of several general practitioners on the basis of the physician's medical records and physiological results of the fasting glycaemia examination. The selection criteria in both diabetics and non-diabetics respected age and gender-proportion comparability among groups; all subjects derived from the same geographical area and all were of the same ethnic (Caucasian) origin. Informed consent was obtained from all patients prior to their inclusion in the study. The study was performed in accordance with the principles of the Declaration of Helsinki and approved by the Committee for Ethics of Medical Experiments on Human Subjects, Medical Faculty, Masaryk University Brno.

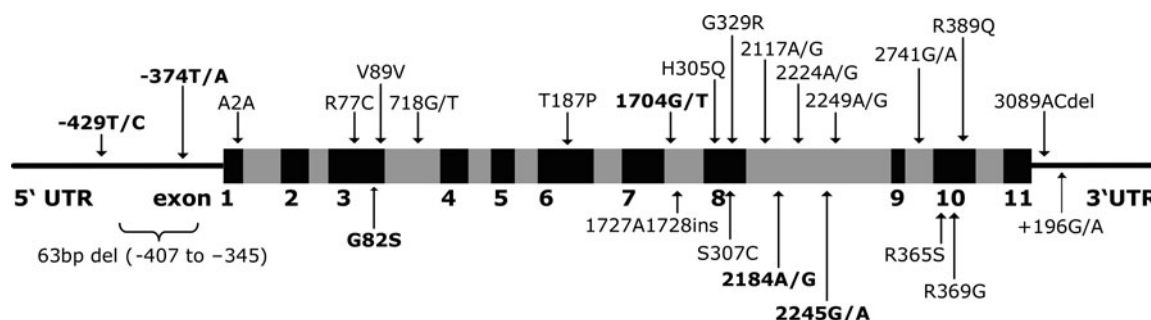


Fig. 1. SNP map of the RAGE gene. Common variants marked in bold.

Table 1. Clinical characteristics of the subjects

Parameter	DM + DN (<i>n</i> = 198)	DM non-DN (<i>n</i> = 179)	Non-DM (<i>n</i> = 228)
Age (years)	63.8 (±13.3)	63.1 (±11.1)	59.1 (±15.5)
DM duration before DN onset (years)	11.1 (±8.4)	5.5 (±6.3) ^a	—
Hypertension (%)	86.4	63.1	38.6
Micro _{com} (mean/median)	1.72/2	1.56/1	—
Macro _{com} (mean/median)	0.96/1	0.74/0	0.69/1
FPG (mmol/l)	8.88 (±3.84)	7.02 (±2.59)	4.97 (±0.72)
HbA _{1c} (%)	7.20 (±2.18)	6.25 (±1.63)	4.67 (±0.77)
Urea (μmol/l)	14.60 (±7.77)	6.40 (±1.90)	5.78 (±1.37)
Creatinine(μmol/l)	220.33 (±175.13)	88.36 (±14.94)	79.56 (±12.89)
AER (g/24 h)	1.12 (±1.58)	—	—
GFR (ml/s)	0.81 (±0.41)	—	—

If not stated differently, data are expressed as means (±SD).

^aIn the DM non-DN group the duration of diabetes from the time of diagnosis until recruitment to the study is reported. Micro_{com}, microvascular comorbidity summary index, presence of both diabetic retinopathy and/or neuropathy contributes 1 to the index; Macro_{com}, macrovascular comorbidity summary index, presence of CHD and/or peripheral vascular disease and/or cerebrovascular disease contributes 1 to the index.

Diagnosis of diabetic nephropathy

The presence of renal complications was assessed on the basis of the patients' medical history of diabetes (type 2 in this study) with established antidiabetic treatment, medical record from a nephrologist and recent measurement of albumin excretion rate (AER) and glomerular filtration rate (GFR) at the time of enrolment into this study. According to DN clinical criteria based on GFR and AER (or overt proteinuria), subjects with minimally incipient DN (e.g. persistent microalbuminuria) were indicated for regular follow-up in the nephrology unit and, thus, likely to be included in the association study. Diagnosis of DN for the purpose of this study was based solely on the repeated assessment of AER from the 24 h urine collection (used for simultaneous measurement of creatinine clearance) prior to a visit to a nephrologist. A cut-off value of AER ≥30 mg/24 h (detected repeatedly) was used as an inclusion criterion to the DM + DN group. However, for the purpose of survival analysis, more-stringent criteria were applied and only those with unequivocal persistent proteinuria (AER ≥300 g/24 h) were analysed. Negative results of periodical annual screening for microalbuminuria and normal plasma creatinine in the presence of type 2 diabetes were used as criteria for DM non-DN subjects.

Diagnosis of other micro- and macrovascular complications

The presence of diabetic complications other than DN (micro- and macrovascular) was assessed in both DM + DN and DM non-DN groups by summary comorbidity indexes (micro_{com} and macro_{com}, respectively), where presence of both diabetic retinopathy and/or neuropathy and presence of coronary heart disease (CHD) and/or peripheral vascular disease and/or cerebrovascular disease, respectively, contributed a value of 1 to the index (Table 1). Presence of diabetic retinopathy was diagnosed by means of direct ophthalmoscopy. Peripheral and/or autonomic diabetic neuropathy was diagnosed according to the presence of particular symptoms and results of a physical examination. Diagnosis of CHD was based on the existence of angina

pectoris or previous history of myocardial infarction. Symptoms of claudication and/or presence or patient's history of trophic ulcerations of lower extremities were decisive for the diagnosis of peripheral vascular disease. Cerebrovascular disease was diagnosed according to patient's history of stroke, partial reversible ischaemic deficit or transitory ischaemic attacks.

Detection of polymorphisms and construction of haplotypes

DNA was isolated from peripheral blood leukocytes by a standard extraction method and was available for all subjects enrolled in the study. From all described RAGE substitutions, only those with minor allele frequency (MAF) ≥1% were studied: −429T/C, −374T/A, G82S, 1704G/T, 2184A/G and 2245G/A. Promoter SNPs −429T/C and −374T/A were genotyped together with the rare −407 to −345 63-bp insertion/deletion from a single polymerase chain reaction (PCR) digested by a mixture of restriction endonucleases. Since this method detects true promoter haplotypes, information about partial phasing could be used in subsequent haplotype construction and improve such estimation. Genotyping was performed according to the methods described elsewhere [10,12,16]. Linkage Disequilibrium Analyzer 1.0 (<http://www.chgb.org.cn/lda/lda.htm>) [17] was used for analysis of linkage disequilibrium (LD); the pairwise LD measure *D'* was used to describe LD among SNPs.

PHASE v. 2.0 [18,19] and HAPLOTYPYER [20] software were used to resolve a sample of phase-unknown multilocus genotypes and to estimate population haplotype frequencies. Due to reported selective advantages of these two Bayesian-based algorithms in some special cases — robustness towards departure from Hardy–Weinberg equilibrium (HWE) (HAPLOTYPYER), handling partially experimentally phased subjects and reflecting relative distances between markers (PHASE) — resolving of haplotypes was performed by both algorithms and results were compared. SNPtagger (<http://www.well.ox.ac.uk/~xiayi/haplotype/index.html>) [21] and HapBlock (<http://www.cmb.usc.edu/~msms/HapBlock>) [22] programs were used to select htSNPs.

Statistical analysis

Differences in genotype distributions from those expected for HWE were tested by a χ^2 test. Differences in allele frequencies of SNPs fulfilling HWE expectation were tested by a two-tail Fisher exact test; otherwise, a χ^2 test for genotype frequencies was used. Comparison of estimated haplotype frequencies was performed as described [23] by (i) conducting separate one-degree of freedom (df) tests for a series of 2×2 contingency tables testing the frequency of each specific haplotype *vs* all others between the two groups (two-tail Fisher exact test) and (ii) omnibus testing of differences in haplotype frequency profiles between the two groups (statistical significance assessed empirically via permutation testing). $P < 0.05$ was considered statistically significant. Testing of allele or genotype frequencies for several SNPs or series of contingency tables for haplotypes among the three groups involves multiple comparisons; therefore, necessary correction had to be applied. Where appropriate, Bonferroni correction was used to adjust the α level according to the number of independent comparisons to an overall value of 0.05. Adjusted P -values for particular analyses are denoted as P_{corr} . Additionally, haplotype-specific effects were analysed by logistic regression-based algorithm, as described by Zhao *et al.* [24].

Survival analysis in a group of diabetic patients was performed using the Kaplan–Meier method and Cox's proportional hazard model. The aim of the survival analysis was to retrospectively analyse the relationship between haplotype and duration of DN (assessed according to personal history of DN in each diabetic). Kaplan–Meier survival curves were constructed for three RAGE₂ haplotype groups (homozygous RAGE₂ +/+, heterozygous RAGE₂ +/- and homozygous RAGE₂ -/-). Subjects with DN (subgroup of DM + DN with AER ≥ 300 g/24 h) were classified as complete responses, others as censored; survival represented duration of DM until the onset of DN in complete responses or DM duration until the recruitment to the study in censored responses. Survival was compared among multiple groups and post-hoc comparisons were made between pairs of groups (log rank test).

Variation of selected biochemical and clinical parameters corresponding to particular genotypes was tested by Kruskal–Wallis analysis of variance (ANOVA) and Spearman correlation coefficient. For all statistical analyses, STATISTICA for Windows® (Statsoft Inc., Tulsa, OK, USA) was used.

Results

Genotype and allele frequencies of all polymorphisms studied are given in Table 2. Data analysis employed the following strategies: (i) single-locus analyses of genotype data and LD assessment; (ii) *in silico* inferring of haplotypes and comparison of their frequencies among groups; (iii) analysis of duration data by survival analysis; and (iv) analysis of genetic data in relation to the clinical and biochemical phenotype.

Single-locus analyses

After the correction for multiple comparisons, there were neither significant deviations from HWE nor any

significant differences in allele or genotype frequencies between groups (Table 2). Frequencies of the -429C and 2184G alleles were, however, higher in the DN group (22.0% *vs* average 16.3% in control groups and 23.0% *vs* average 15.2%, respectively). Allele frequencies of the 63 bp insertion/deletion polymorphism obtained as a by-product of promoter genotyping were 99.2% insertion, 0.8% deletion in both DM + DN and DM non-DN and 99.6% insertion, 0.4% deletion in non-DM.

Figure 2 displays results of LD assessment in the whole sample and the three groups separately. The pairwise LD values described by Lewontin's D' suggested that most of the SNPs are in strong LD. However, some SNPs (namely the G82S) showed relatively low LD with markers of suggestive association with DN (-429T/C and 2184A/G), see arrows in Figure 2. Low LD together with the low frequency of G82S might suggest distant origin and possible independency of this SNP on common haplotypes in the RAGE gene. Alternatively, low LD could be due to a high mutation rate in the particular gene region. However, there are no indications that the G82S is localized in a mutation hotspot.

In silico haplotyping and haplotype analysis

Experimental phasing (molecular haplotyping) of all individuals was only available for 5'UTR polymorphisms detected by allele-specific PCR (-429T/C and -374T/A). However, for inferring and comparing haplotype frequencies we theoretically considered the whole genotype set as unphased and promoter haplotypes were used only for the control purposes and for reverse classification of subjects for survival analysis (the - k option in PHASE v. 2.0). Inferring haplotypes from unphased genotype data was performed computationally using HAPLOTYPER and PHASE v. 2.0 software packages with the assumption of mutual independence of haplotypes. Implementing the information about partial phasing of individuals and performing the case-control comparison simultaneously might make PHASE the algorithm of choice. Nevertheless, *in silico* haplotyping was performed by both types of software. Practically, PHASE was run with a single file including genotypes of all subjects; for c-option (three groups), 10 000 permutations were performed. HAPLOTYPER was run with 30 rounds on each group of subjects and on pooled case-control groups (due to the limitation of maximal allowed number of subjects). Completeness of genotype data was nearly 100% (total of five missing genotypes). Table 3 shows haplotype frequency estimates in particular groups. Haplotypes were denoted RAGE_x where x represents ordinal number of inferred haplotype (total $x = 10$ in DM + DN group; $x = 11$ in the other two groups). Using permutation testing (PHASE, 10 000 permutations), we found significant omnibus differences in haplotype frequencies between DM + DN *vs* DM non-DN groups and DM + DN *vs* non-DM groups (both $P = 0.0007$ and $P = 0.0013$, respectively).

Table 2. Genotype and allele frequencies of RAGE polymorphisms in the three groups. Comparisons performed by means of two-tail Fisher-exact test, for the 2184A/G (violated HWE in DM + DN group) χ^2 test was used. Bonferroni corrected *P*-value for single test (12 independent comparisons) equals to 0.004

	DM + DN (<i>n</i> = 198)						DM non-DN (<i>n</i> = 179)						non-DM (<i>n</i> = 228)					
	TT	TC	CC	T (%)	C (%)		TT	TC	CC	T (%)	C (%)		TT	TC	CC	T (%)	C (%)	
−429T/C	125	59	14 ^b	78.0	22.0		129	46	4	84.9	15.1		158	60	9	82.8	17.2	
							<i>P</i> = 0.019 (<i>P</i> _{corr} = NS)						<i>P</i> = NS					
−374T/A ^a	82	92	21	65.6	34.4		65	90	21	62.5	37.5		90	113	22	65.1	34.9	
							<i>P</i> = NS						<i>P</i> = NS					
G82S	GG 192	GS 6	SS 0	G (%) 98.5	S (%) 1.5		GG 166	GS 13	SS 0	G (%) 96.4	S (%) 3.6		GG 223	GS 5	SS 0	G (%) 98.9	S (%) 1.1	
							<i>P</i> = NS						<i>P</i> = NS					
1704G/T	GG 173	GT 22	TT 1	G (%) 93.9	T (%) 6.1		GG 159	GT 20	TT 0	G (%) 94.4	T (%) 5.6		GG 207	GT 20	TT 1	G (%) 95.2	T (%) 4.8	
							<i>P</i> = NS						<i>P</i> = NS					
2184A/G	AA 124	AG 57	GG 17 ^b	A (%) 77.0	G (%) 23.0		AA 131	AG 43	GG 5	A (%) 85.2	G (%) 14.8		AA 164	AG 57	GG 7	A (%) 84.4	G (%) 15.6	
							<i>P</i> = 0.021 (<i>P</i> _{corr} = NS)						<i>P</i> = 0.022 (<i>P</i> _{corr} = NS)					
2245G/A	GG 142	GA 51	AA 4	G (%) 85.0	A (%) 15.0		GG 124	GA 50	AA 5	G (%) 83.2	A (%) 16.8		GG 179	GA 47	AA 2	G (%) 88.8	A (%) 11.2	
							<i>P</i> = NS						<i>P</i> = NS					

^aIn the presence of a deletion allele of the rare 63 bp I/D variant, genotypes of −374T/A consist of either T/− or A/−. Total of 8 such cases detected in the whole set (not shown in this table), all included into further haplotype analyses; ^bdistribution of genotypes marginally deviate from Hardy–Weinberg equilibrium (*P*_{−429T/C} = 0.065, *P*_{2184A/G} = 0.0086; total of 18 tests, *P*_{corr} = NS for both SNPs).

Differences among the two control groups were not found. RAGE₂ (CTGGGG) and RAGE₈ (CTGGAG) haplotype frequencies accounted for the greatest part of the difference in both comparisons (all $P_{\text{corr}} < 0.003$). Particularly, the frequency of the RAGE₂ in the DM + DN group was markedly higher than in the two control groups (21.7% vs 12.8% and 13.8%, respectively). The frequency of RAGE₈ in the DM + DN group was, conversely, lower than expected; this haplotype had, however, an incomparably lower population frequency than common haplotype RAGE₂ and its eventual effect is, therefore, likely to be minor.

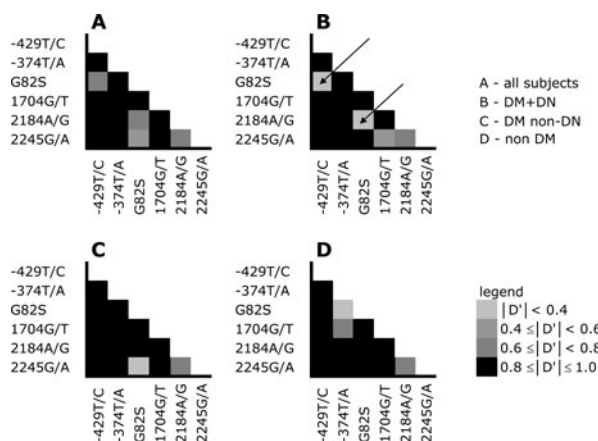


Fig. 2. Plot of pairwise linkage disequilibrium (D') values for the six SNPs studied. $|D'|$ values shaded in greyscale according to the legend. (A) Whole set of subjects, (B) DM + DN group, (C) DM non-DN group and (D) non-DN group. Arrows indicate a SNP (G82S) with a relatively low LD with markers of suggestive association with DN (–429T/C and 2184A/G).

In addition to the global test for haplotype association, we also estimated the haplotype-specific effects of each haplotype on disease risk. The latter was performed by the method described by Zhao *et al.* [24], which allows for joint estimation of haplotype distribution and its correlation with disease phenotype via logistic regression. The best reconstructed haplotypes are treated as covariates in logistic regression analysis. Estimated odds ratios (ORs) and 95% confidence intervals (CIs) for common haplotypes are shown in Table 3. Comparison of the DM + DN group with the DM non-DN and the non-DM group revealed for RAGE₂ haplotype ORs 1.65 (95% CI: 1.08–2.50; $P = 0.020$) and 1.79 (95% CI: 1.22–2.62; $P = 0.003$), respectively [for pooled control groups, OR = 1.72 (95% CI: 1.22–2.42; $P = 0.002$)], indicating again a significant increase of relative risk of DN associated with this haplotype. We therefore conclude that this common haplotype, associated with DN, could be regarded as a marker for DN.

Identification of htSNPs

Substitutions at positions –429 and, especially, 2184 are obviously htSNPs for the common haplotype RAGE₂. To select efficient htSNPs in the RAGE gene more rigorously, we applied two algorithms proposed by Ke and Cardon [21] and Zhang *et al.* [22]. With both programs, htSNPs were selected to explain an interval of 100–50% of the haplotype diversity given the SNPs analysed. Results are displayed in Table 4. Depending on the particular study requirements, one can ignore uncommon haplotypes by setting a haplotype ‘coverage value’ less than 1.0.

Table 3. Multilocus haplotype frequency estimates and results of comparison of estimated haplotype frequencies. Haplotypes are ordered according to decreasing haplotype frequency in the DM + DN group, those with frequencies <1% in all three groups were pooled together as ‘others’. Estimates of frequencies given in absolute numbers (according to HAPLOTYPYER), percentages in parentheses (HAPLOTYPYER/PHASE). Separate one-degree of freedom tests were conducted for a series of 2×2 contingency tables testing the frequency of each specific haplotype vs all others between particular groups (two-tail Fisher exact test, 20 independent observations, Bonferroni-corrected P -value for single test equals to 0.003). ORs and 95% CIs for common haplotypes were estimated by logistic regression analysis, treating the best-reconstructed haplotypes as covariates. Global differences in haplotype frequency profiles between particular groups were tested by permutation testing (PHASE output, 10 000 permutations)

RAGE _x	Haplotype	DM + DN	DM non-DN	χ^2 (^a P-value)	OR (95% CI)	Non-DM	χ^2 (^a P-value)	OR [95% CI]
1	TTGGAG	135 (34.1/34.4)	129 (36.0/35.6)	NS	1.00 [reference]	185 (40.6/40.5)	NS	1.00 [reference]
2	CTGGGG	86 (21.7/21.7)	46 (12.8/12.8)	10.24 (0.0015)	1.65 [1.08–2.50] ^b	63 (13.8/13.8)	9.17 (0.0028)	1.79 [1.22–2.62] ^b
3	TAGGAG	81 (20.5/20.5)	76 (21.2/21.9)	NS	0.94 [0.63–1.42]	109 (23.9/23.9)	NS	1.09 [0.74–1.62]
4	TAGGAA	54 (13.6/13.6)	56 (15.6/15.3)	NS	0.94 [0.59–1.49]	47 (10.3/10.4)	NS	1.55 [0.97–2.48]
5	TTGTAG	25 (6.3/6.0)	20 (5.6/5.5)	NS	1.09 [0.57–2.08]	21 (4.6/4.6)	NS	1.50 [0.79–2.83]
6	TTSAG	5 (1.3/1.2)	13 (3.6/3.6)	NS	0.40 [0.13–1.22]	5 (1.1/1.1)	NS	–
7	TTGGAA	4 (1.0/1.2)	3 (0.8/1.0)	NS	–	2 (0.4/0.5)	NS	–
8	CTGGAG	0 (0.0/0.0)	8 (2.2/2.2)	8.94 (0.0025)	–	15 (3.3/3.3)	13.26 (0.000085)	–
9	TTGGGG	3 (0.8/0.8)	3 (0.8/1.2)	NS	–	6 (1.3/1.2)	NS	–
10	Others	3 (0.7/0.6)	4 (1.4/0.9)	NS	–	3 (0.7/0.7)	NS	–
		(1) vs. (2)		^c P=0.0007	(1) vs. (3)		^c P=0.0013	

^a P -value derived from testing of single 2×2 contingency tables.

^bIndicates that the ORs are different from 1 at a significance level of <0.05.

^cOmnibus P -value assessed empirically by permutation testing.

Table 4. Comparison of the results of RAGE htSNP selection by two different methods. Numbers denote SNPs according to their position in the gene (from 5' to 3' orientation) and in Table 2. Alternative markers without additional information shown in brackets

	Coverage of the haplotype diversity					
	1.0	0.9–1.0	0.8–0.9	0.7–0.8	0.6–0.7	0.5–0.6
SNPtagger	All SNPs	1(5), 2, 4, 6	1(5), 2, 6	1(5), 2	1(5), 2	2
HapBlock ^a	All SNPs	1(5), 2, 4, 6	1(5), 2, 6	1(5), 2	1(5), 2	2

^aParameters for the selection of htSNPs in HapBlock were set as follows: (i) fixed genome coverage with the minimum number of tag SNPs for block partition, (ii) haplotypes inferred from genotypes by EM-algorithm served as an input data, (iii) block definition was based on coverage of common haplotypes and (iv) definition of tag SNPs was based on haplotype diversity.

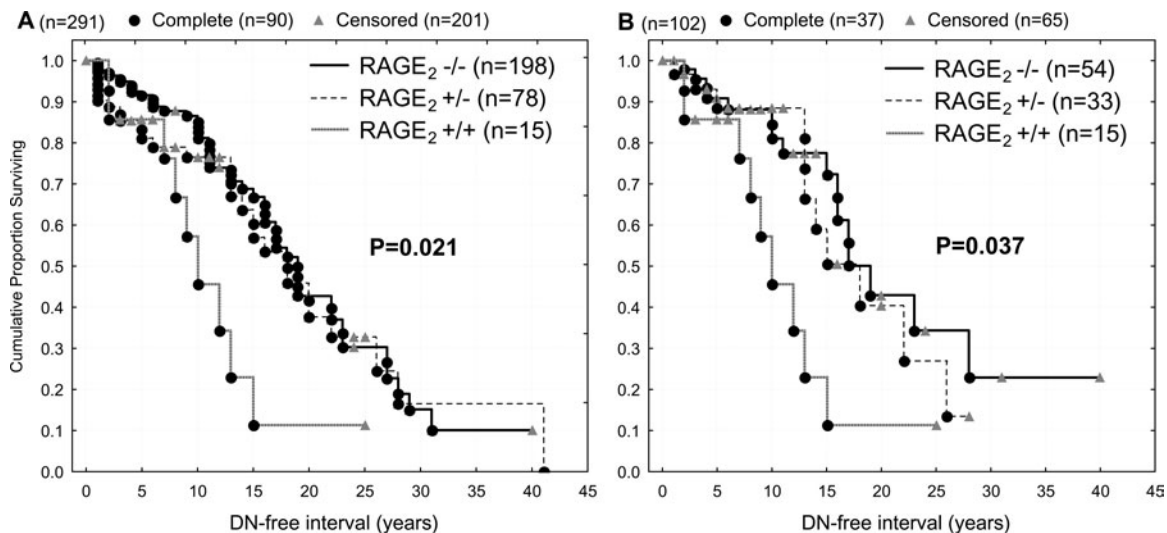


Fig. 3. Results of survival analysis. Kaplan–Meier curves for the RAGE₂ diplotype groups. Diabetic subjects with AER ≥ 300 g/24 h classified as complete responses, others as censored. DN-free interval ('survival') represents duration of DM till the onset of DN in complete responses or DM duration till the recruitment to the study in censored responses. RAGE₂ diplotypes were classified on the basis of PHASE estimates as the most probable haplotype pairs for each individual weighed against their posterior probabilities (for all included subjects $P \geq 0.95$). Comparison A included all available subjects classified for RAGE₂ haplotype, irrespective of the presence of other haplotypes in heterozygotes and non-RAGE₂ homozygotes; comparison B included a subset of subjects bearing in heterozygote and wild-type homozygote diplotypes only RAGE₁ haplotypes or those containing rare variants (MAF <10%) in combination with RAGE₂. Reported *P*-values apply to comparison of the DN-free interval among the three RAGE₂ diplotype groups.

Survival analysis

Using the Kaplan–Meier method, survival functions were constructed for three groups of subjects classified according to the RAGE₂ diplotype (carriers of two, one and zero copies of the RAGE₂ haplotype). Naturally, unambiguous reverse classification of estimated haplotypes into real genotypes in each individual subject was not possible; nevertheless, with the help of estimates of the most probable haplotype pairs for each individual produced by PHASE, weighed against their posterior probabilities, survival analysis was performed in a subset of diabetic subjects where classification of the RAGE₂ genotype was sufficiently reliable. Presence of two (+/+) or zero copies (–/–) of the RAGE₂ haplotype was clearly distinguishable. For compound heterozygotes –429T/C, 2184A/G, minor alleles in the RAGE₂ haplotype could theoretically be mutually in either *cis* or *trans* orientation (e.g. C-G/T-A or C-A/T-G, respectively). However, the *trans* position was

not once estimated computationally in the whole sample; therefore, we assumed solely *cis* orientation (e.g. C-G/T-A combination only) accordingly. Survival analysis was performed twice: first (A), including all available subjects classified for the number of RAGE₂ copies (+/+, +/- and -/-), irrespective of the presence of other haplotypes in heterozygotes and non-RAGE₂ homozygotes; and, second (B), in a subset of subjects bearing only RAGE₁ haplotypes or those with variants with MAF <10% in combination with RAGE₂ in heterozygotes and -/- homozygotes. Analysis B should exclude possible interfering effects of other common haplotypes on survival.

Practically, from the total number of diabetic subjects ($n = 377$), those with persistent proteinuria (AER ≥ 300 g/24 h) were classified as complete responses ($n = 99$) and others as censored ($n = 278$). Reliable reverse classification of RAGE₂ haplotypes into genotypes was reliable in all subjects (posterior probabilities for both individual's estimated haplotypes >95%)

and was performed as described above. Data required for survival analysis (DN-free interval) were available in 90 complete responses (duration of diabetes till the onset of persistent proteinuria) and 201 censored (DM duration till the recruitment to the study). Figure 3 shows the Kaplan–Meier plots of survival functions for the three RAGE₂ diplotype groups in the larger (A) and smaller (B) subsets of subjects. The duration of diabetes until the onset of DN, e.g. DN-free interval, was significantly different among RAGE₂ diplotype groups in both A and B comparisons ($\chi^2=7.67$, $df=2$, $P=0.021$ and $\chi^2=6.61$, $df=2$, $P=0.037$, respectively). The median DN-free intervals in A (and B) comparisons were 9.6 (9.6) years in RAGE₂ $+/+$ homozygotes, 17.8 (15.2) years in $+/-$ heterozygotes and 18.9 (17.0) years in the $-/-$ combination. Post-hoc testing revealed a significant difference between RAGE₂ $+/+$ vs $-/-$ and $+/+$ vs $+/-$ groups in both A and B comparisons (all $P<0.05$; log rank test). RAGE₂ $+/-$ vs $-/-$ did not differ significantly. Survival time was a function of neither age nor gender in any genotype group.

Analysis of genetic data in relation to the clinical and biochemical phenotype

Neither micro- nor macrovascular comorbidity (micro_{com} and macro_{com} indexes) was significantly different among particular RAGE₂ haplotype groups tested in the two diabetic groups separately as well as in the pooled group (Spearman correlation coefficient). Similarly, urea, creatinine, proteinuria and GFR did not differ among particular RAGE₂ groups (Kruskal–Wallis ANOVA).

Discussion

Glycooxidation represents one of the principal pathogenic mechanisms of hyperglycaemia-induced cell damage in both types of diabetes. Hyperglycaemia substantially enhances oxidative stress and the AGE/RAGE system is one of the most important non-mitochondrial contributors. AGEs generate ROS directly or through their receptors, presumably by intracellular cooperation of RAGE with membrane-bound NAD(P)H oxidase [25]. An abundance of AGE ligands in the diabetic milieu and the unequivocal role of RAGE in the pathogenesis of DN, together with the identification of extensive polymorphism within its coding and non-coding regions, logically focused attention on the study of association of genetic variability in the RAGE gene with diabetic complications.

Family clustering, apparent heterogeneity in the onset and progression of DN in both type 1 and 2 diabetes and the results of segregation studies indicate that genetic factors contribute to susceptibility to DN. Several linkage studies attempted to find the susceptibility locus for DN. A locus in chromosome 6p21.3, where RAGE lies, has not been identified as one of

them. However, the majority of linkage studies were performed with a prior candidate gene hypothesis; therefore, they were not truly genome-wide. Besides, linkage studies in type 2 diabetes encounter substantial problems arising from late manifestation of disease. The possibility that RAGE is not only a functional but also a positional candidate for DN could not, therefore, be ruled out.

Complex disease gene mapping via case-control studies faces several methodological and statistical considerations. Studies of individual SNPs gene-by-gene are a widely used, simple, easy to interpret approach and certainly legitimate in the initial stage, when it could identify variants with strong marginal effects. In studies where association of more than one SNP is investigated, correction for multiple comparisons causes a poor outcome, because SNPs without informative value worsen results. At the same time, tightly linked SNPs are incorrectly treated as independent. Thus, analysis of haplotypes is a theoretically better approach; however, unambiguous reconstruction of haplotypes from genotype data is not possible (for multiple heterozygotes) and, consequently, eventual reverse classification into haplotype-based diplotypes is also more or less approximate, based on haplotype estimates and posterior probabilities. Due to these reasons, haplotype analyses are likely to slightly overstate eventual statistical significance.

Due to the limited power of conventional single-locus analysis and its logical shortcomings in the case of several linked SNPs within one gene, but aware of the above mentioned inaccuracy, we preferred a 'haplotype-centric' approach. In the present study, we examined the relationship between the genetic variability in the RAGE gene and DN accompanying type 2 diabetes in a Caucasian population. We tried to preserve as much of the SNP information content as possible without immoderate decrease of statistical power. Our results demonstrate first of all (i) significantly increased prevalence of RAGE₂ haplotype (consisting of alleles $-429C/-374T/82G/1704G/2184G/2245G$) in the DN group (21.7% in DN vs 13.4% in control groups) in the situation of (ii) statistically formally non-significant single-locus association of the $-429T/C$ and $2184A/G$ SNPs. Moreover, (iii) the RAGE₂ haplotype was not only associated with DN as such [with relative risk associated with RAGE₂ haplotype corresponding to an OR=1.72 (95% CI: 1.22–2.42; $P=0.002$)], but its presence was significantly associated with acceleration of the onset of DN with the biological gradient corresponding to the number of RAGE₂ copies in each individual group. However, this effect did not correspond with classical clinical markers of DN. Proteinuria and decline of renal function are probably net results of a series of events implicated in the pathogenesis of DN rather than relevant intermediate phenotype for the gene studied.

Ascertained haplotype association could either indicate presence of the true causal variant within the

haplotype or indirectly reflect existence of a causal marker located in the vicinity of the haplotype (in the same but probably another gene). Although the latter is usually the case, available functional data about RAGE are in favour of the hypothesis that the ascertained association might be causal. Allele T of the -429T/C polymorphism was shown to have a 2-fold transcriptional activity compared with the C allele [10] and the 2184A/G substitution in intron 8 was associated with antioxidant status in diabetics [12]. Of particular interest is the finding that not a single association of any of the two SNPs separately but joint occurrence in the RAGE allele contributed to the DN susceptibility in our study. *Cis*-interaction between the two variants might be one plausible explanation. In the light of recent findings of the existence of tissue-specific alternative splicing of RAGE mRNA leading to the endogenous production of N- and C-truncated forms of RAGE [26–28], it might be possible that *cis*-interaction between promoter and intron 8 variant participate in the regulation of that process. It was shown that alternative splicing involves areas spanning exon 10 and introns 6 and 9 of the RAGE gene; htSNP 2184A/G in intron 8 could be hypothetically situated within the regulatory element binding site and participate in a regulatory role. Competition of translational products of the full-length RAGE transcript with its soluble alternative transcript for the AGEs could determine the extent of distal effects following ligand binding. Summarily, the RAGE₂ haplotype could represent a ‘gain-of-function’ variant, whose presence in a particular diabetic subject, especially in homozygous form, contributes to the susceptibility to develop DN. Judging by its fairly common population frequency, we might speculate that this haplotype could bring some selective advantage (prompt innate immune response, for example). However, in agreement with Yonekura *et al.* [28], ‘diabetes abuses the molecular devices of the RAGE signalling pathway primarily evolved for other physiological processes’ and turns them into pathogenic factors.

Several limitations of our study have to be mentioned. First, the number of subjects was not sufficiently large enough to detect an eventual association between DN and polymorphisms with lower frequencies (G82S and 1704G/T); power analysis (data not shown) was conducted to estimate a sample size large enough to detect differences in allele frequencies of SNPs with $\text{MAF} \geq 10\%$ (according to published frequencies by Kankova *et al.* [16] and Hudson *et al.* [10]). However, single-locus analysis has not been our primary goal and, besides, such a low population frequency does not correspond with those expected for the proposed model of inheritance. Second, the control group of diabetics without DN had, on average, a shorter duration of diabetes. This is a serious argument potentially weakening our conclusions, as one could expect future development of DN at least in some of the control subjects. However, with respect to the equal mean age in both groups, genetic similarity of the DM non-DN group

with non-diabetics and a rather random diagnosis of type 2 diabetes, we could hypothesize that either DM developed later in life or was clinically silent for a longer time and, thus, undiagnosed because those subjects do not belong to the high-risk group. For control purposes, all analyses mentioned above were performed on the subset of DM + DN with duration of DM < 10 years and DM non-DN with duration > 10 years yielding analogous results to those for the whole diabetic group with the exception of even more pronounced HWE violation for the -429T/C in DN group (data not shown). Nevertheless, this issue deserves further attention and recruitment of a large control group of subjects with long duration of diabetes without chronic complications is currently ongoing.

In conclusion, the presented data indicate that the RAGE₂ haplotype containing minor alleles at positions -429 and 2184 and major allele at position -374 could be regarded as a marker for DN; consequently, variant -374T/A plus one of the two SNPs -429T/C or 2184A/G could be considered htSNPs for the RAGE gene covering a large proportion of haplotype diversity and used in future studies examining association of RAGE with disease phenotypes. Experimental evidence of the RAGE pathogenic involvement, consistency of association of genetic variability in RAGE with DN (allelic heterogeneity in different ethnic groups, though) and the observed negative correlation of the DN-free interval and the number of RAGE₂ haplotype copies support the hypothesis that the reported association could be causal.

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Conflict of interest statement. None declared.

References

1. Heidland A, Sebekova K, Schinzel R. Advanced glycation end products and the progressive course of renal disease. *Am J Kidney Dis* 2001; 38: S100–S106
2. Forbes JM, Cooper ME, Oldfield MD, Thomas MC. Role of advanced glycation end products in diabetic nephropathy. *J Am Soc Nephrol* 2003; 14: S254–S258
3. Bierhaus A, Schiekofer S, Schwaninger M *et al.* Diabetes-associated sustained activation of the transcription factor nuclear factor-kappaB. *Diabetes* 2001; 50: 2792–2808
4. Tanji N, Markowitz GS, Fu C *et al.* Expression of advanced glycation end products and their cellular receptor RAGE in diabetic nephropathy and nondiabetic renal disease. *J Am Soc Nephrol* 2000; 11: 1656–1666
5. Morcos M, Sayed AA, Bierhaus A *et al.* Activation of tubular epithelial cells in diabetic nephropathy. *Diabetes* 2002; 51: 3532–3544
6. Wendt TM, Tanji N, Guo J *et al.* RAGE drives the development of glomerulosclerosis and implicates podocyte activation in the

- pathogenesis of diabetic nephropathy. *Am J Pathol* 2003; 162: 1123–1137
7. Bonnardel-Phu E, Wautier JL, Schmidt AM, Avila C, Vicaute E. Acute modulation of albumin microvascular leakage by advanced glycation end products in microcirculation of diabetic rats *in vivo*. *Diabetes* 1999; 48: 2052–2058
 8. Yamamoto Y, Kato I, Doi T *et al.* Development and prevention of advanced diabetic nephropathy in RAGE-overexpressing mice. *J Clin Invest* 2001; 108: 261–268
 9. Jerums G, Panagiotopoulos S, Forbes J, Osicka T, Cooper M. Evolving concepts in advanced glycation, diabetic nephropathy, and diabetic vascular disease. *Arch Biochem Biophys* 2003; 419: 55–62
 10. Hudson BI, Stickland MH, Futers TS, Grant PJ. Effects of novel polymorphisms in the RAGE gene on transcriptional regulation and their association with diabetic retinopathy. *Diabetes* 2001; 50: 1505–1511
 11. Hofmann MA, Drury S, Hudson BI *et al.* RAGE and arthritis: the G82S polymorphism amplifies the inflammatory response. *Genes Immun* 2002; 3: 123–135
 12. Kankova K, Marova I, Zahejsky J *et al.* Polymorphisms 1704G/T and 2184A/G in the RAGE gene are associated with antioxidant status. *Metabolism* 2001; 50: 1152–1160
 13. Pettersson-Fernholm K, Forsblom C, Hudson BI, Perola M, Grant PJ, Groop PH. The functional –374 T/A RAGE gene polymorphism is associated with proteinuria and cardiovascular disease in type 1 diabetic patients. *Diabetes* 2003; 52: 891–894
 14. Matsunaga-Irie S, Maruyama T, Yamamoto Y *et al.* Relation between development of nephropathy and the p22phox C242T and receptor for advanced glycation end product G1704T gene polymorphisms in type 2 diabetic patients. *Diabetes Care* 2004; 27: 303–307
 15. Rudofsky G Jr, Isermann B, Schilling T *et al.* A 63bp deletion in the promoter of RAGE correlates with a decreased risk for nephropathy in patients with type 2 diabetes. *Exp Clin Endocrinol Diabetes* 2004; 112: 135–141
 16. Kankova K, Zahejsky J, Marova I *et al.* Polymorphisms in the RAGE gene influence susceptibility to diabetes-associated microvascular dermatoses in NIDDM. *J Diabetes Complic* 2001; 15: 185–192
 17. Ding K, Zhou K, He F, Shen Y. LDA – a Java-based linkage disequilibrium analyzer. *Bioinformatics* 2003; 19: 2147–2148
 18. Stephens M, Smith NJ, Donnelly P. A new statistical method for haplotype reconstruction from population data. *Am J Hum Genet* 2001; 68: 978–989
 19. Stephens M, Donnelly P. A comparison of Bayesian methods for haplotype reconstruction from population genotype data. *Am J Hum Genet* 2003; 73: 1162–1169
 20. Niu T, Qin ZS, Xu X, Liu JS. Bayesian haplotype inference for multiple linked single-nucleotide polymorphisms. *Am J Hum Genet* 2002; 70: 157–169
 21. Ke X, Cardon LR. Efficient selective screening of haplotype tag SNPs. *Bioinformatics* 2003; 19: 287–288
 22. Zhang K, Qin Z, Chen T, Liu JS, Waterman MS, Sun F. HapBlock: haplotype block partitioning and tag SNP selection software using a set of dynamic programming algorithms. *Bioinformatics* 2005; 21: 131–134
 23. Fallin D, Cohen A, Essioux L *et al.* Genetic analysis of case/control data using estimated haplotype frequencies: application to APOE locus variation and Alzheimer's disease. *Genome Res* 2001; 11: 143–151
 24. Zhao LP, Li SS, Khalid N. A method for the assessment of disease associations with single-nucleotide polymorphism haplotypes and environmental variables in case-control studies. *Am J Hum Genet* 2003; 72: 1231–1250
 25. Wautier MP, Chappey O, Corda S, Stern DM, Schmidt AM, Wautier JL. Activation of NADPH oxidase by AGE links oxidant stress to altered gene expression via RAGE. *Am J Physiol Endocrinol Metab* 2001; 280: E685–E694
 26. Malherbe P, Richards JG, Gaillard H *et al.* cDNA cloning of a novel secreted isoform of the human receptor for advanced glycation end products and characterization of cells co-expressing cell-surface scavenger receptors and Swedish mutant amyloid precursor protein. *Brain Res Mol Brain Res* 1999; 71: 159–170
 27. Schlueter C, Hauke S, Flohr AM, Rogalla P, Bullerdiek J. Tissue-specific expression patterns of the RAGE receptor and its soluble forms — a result of regulated alternative splicing? *Biochim Biophys Acta* 2003; 1630: 1–6
 28. Yonekura H, Yamamoto Y, Sakurai S *et al.* Novel splice variants of the receptor for advanced glycation end-products expressed in human vascular endothelial cells and pericytes, and their putative roles in diabetes-induced vascular injury. *Biochem J* 2003; 370: 1097–1109

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